-- DEMO 3.1 – Внутренние соединения (Inner Join)

```
-- Базовое внутренне соединение
SELECT SalesLT.Product.Name As ProductName, SalesLT.ProductCategory.Name AS Category
FROM SalesLT.Product
INNER JOIN SalesLT.ProductCategory
ON SalesLT.Product.ProductCategoryID = SalesLT.ProductCategory.ProductCategoryID;

-- Псевдонимы для таблиц
SELECT p.Name As ProductName, c.Name AS Category
FROM SalesLT.Product AS p
JOIN SalesLT.ProductCategory As c
ON p.ProductCategoryID = c.ProductCategoryID;

-- Соединение для 3+ таблиц
SELECT oh.OrderDate, oh.SalesOrderNumber, p.Name As ProductName, od.OrderQty, od.UnitPrice,
od.LineTotal
FROM SalesLT.SalesOrderHeader AS oh
JOIN SalesLT.SalesOrderDetail AS od
ON od.SalesOrderID = oh.SalesOrderID
JOIN SalesLT.Product AS p
ON od.ProductID = p.ProductID
ORDER BY oh.OrderDate, oh.SalesOrderID, od.SalesOrderDetailID;

-- Составные предикаты в соединении
SELECT oh.OrderDate, oh.SalesOrderNumber, p.Name As ProductName, od.OrderQty, od.UnitPrice,
od.LineTotal
FROM SalesLT.SalesOrderHeader AS oh
JOIN SalesLT.SalesOrderDetail AS od
ON od.SalesOrderID = oh.SalesOrderID
JOIN SalesLT.Product AS p
ON od.ProductID = p.ProductID AND od.UnitPrice = p.ListPrice – составной предикат
ORDER BY oh.OrderDate, oh.SalesOrderID, od.SalesOrderDetailID;
```

-- DEMO 3.2 – Внешние соединения

```
-- получить ВСЕХ клиентов с заказами (заказы только для тех, кто что-то купил)
SELECT c.FirstName, c.LastName, oh.SalesOrderNumber
FROM SalesLT.Customer AS c
LEFT OUTER JOIN SalesLT.SalesOrderHeader AS oh
ON c.CustomerID = oh.CustomerID
ORDER BY c.CustomerID;

-- вернуть только тех клиентов, у которых нет заказов
SELECT c.FirstName, c.LastName, oh.SalesOrderNumber
FROM SalesLT.Customer AS c
LEFT OUTER JOIN SalesLT.SalesOrderHeader AS oh
ON c.CustomerID = oh.CustomerID
WHERE oh.SalesOrderNumber IS NULL
ORDER BY c.CustomerID;


-- Соединение для 3+ таблиц
SELECT p.Name As ProductName, oh.SalesOrderNumber
FROM SalesLT.Product AS p
LEFT JOIN SalesLT.SalesOrderDetail AS od
ON p.ProductID = od.ProductID
LEFT JOIN SalesLT.SalesOrderHeader AS oh -- дополнительные таблицы, добавляемые справа
должны использовать left join
ON od.SalesOrderID = oh.SalesOrderID
ORDER BY p.ProductID;
```

```sql
SELECT p.Name As ProductName, c.Name AS Category, oh.SalesOrderNumber
FROM SalesLT.Product AS p
LEFT OUTER JOIN SalesLT.SalesOrderDetail AS od
ON p.ProductID = od.ProductID
LEFT OUTER JOIN SalesLT.SalesOrderHeader AS oh
ON od.SalesOrderID = oh.SalesOrderID
INNER JOIN SalesLT.ProductCategory AS c -- добавляем к левой части, поэтому можем
использовать inner join
ON p.ProductCategoryID = c.ProductCategoryID
ORDER BY p.ProductID;
```

-- DEMO 3.3 – Декартово произведение (Cross Join)

```sql
-- позвонить каждому клиенту по каждому товару
SELECT p.Name, c.FirstName, c.LastName, c.Phone
FROM SalesLT.Product as p
CROSS JOIN SalesLT.Customer as c;
```

-- DEMO 3.4 – Самосоединение (Self Join)

```sql
-- для примера создадим табличку Employee
CREATE TABLE SalesLT.Employee
(EmployeeID int IDENTITY PRIMARY KEY,
EmployeeName nvarchar(256),
ManagerID int);
GO
-- получаем продавцов из таблицы Customer и генерируем менеджеров
INSERT INTO SalesLT.Employee (EmployeeName, ManagerID)
SELECT DISTINCT Salesperson, NULLIF(CAST(RIGHT(SalesPerson, 1) as INT), 0)
FROM SalesLT.Customer;
GO
UPDATE SalesLT.Employee
SET ManagerID = (SELECT MIN(EmployeeID) FROM SalesLT.Employee WHERE ManagerID IS NULL)
WHERE ManagerID IS NULL
AND EmployeeID > (SELECT MIN(EmployeeID) FROM SalesLT.Employee WHERE ManagerID IS NULL);
GO

-- Вот сама демка с самосоединением
SELECT e.EmployeeName, m.EmployeeName AS ManagerName
FROM SalesLT.Employee AS e
LEFT JOIN SalesLT.Employee AS m
ON e.ManagerID = m.EmployeeID
ORDER BY e.ManagerID;
```